College of Engineering and Computer Science

COMP4010 - Data Visualization

Project 2

Exploring Growth and Concentration in AI Research

Group 6

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GitHub Repository:
https://github.com/The-platypus20/Project_2_Data_Visualization_Group_6

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1 Project Description

This project explores how AI research has evolved during the rapid expansion of modern artificial intelligence, particularly after the rise of deep learning, generative AI, and large language models. Instead of only measuring publication volume, the project investigates how scientific attention, citation impact, topic concentration, and institutional influence change over time.

The final product will be an interactive Python Shiny dashboard that guides users through four connected analytical views: publication growth, research impact, concentration, and research pressure indicators.

The project aims to answer the following core question:

> *"How is the global AI research growth landscape structurally organized across different subfields, institutions, and geographies, and how have these scientific trajectories shifted over time?"*

2 Motivation

AI research is expanding too fast for researchers, policymakers, and industry analysts to track manually. This project is motivated by the need to provide a comprehensive, interactive map of the AI ecosystem, helping users identify not just how much AI research is growing, but where the technical hot spots are, who is leading them, and whether scientific attention is diversifying or intensifying around a few dominant sectors.

3 Dataset Description

The primary dataset will be collected from the OpenAlex API, an open scholarly database containing metadata on academic publications, citations, authors, institutions, venues, and research topics. The current sample dataset contains 7,440 AI-related scholarly works published between 2000 and 2026, with plans to expand to around 20,000 - 50,000 papers using broader AI-related search terms such as "machine learning", "deep learning" and "large language model".

The dataset comprises several interconnected dimensions, structured to support multi-layered exploration:

- **Publication Metadata & Impact Metrics:** Paper title, publication year, venue/source, citation counts, and forward/backward citation links to measure scientific attention and citation inequality.
- **Hierarchical Research Context (Topics & Fields):** Instead of relying on noisy, unstandardized raw keywords, the project leverages OpenAlex's structured **Concept/Topic Taxonomy**. This hierarchical system allows for multi-level filtering and drill-down analysis:
  - *Level 0 & 1 (Macro Domain):* Broad fields such as Computer Science and Artificial Intelligence.
  - *Level 2 & 3 (Subfields & Methods):* Specialized sub-domains like Deep Learning, Natural Language Processing (NLP), Computer Vision, and Reinforcement Learning. This structured taxonomy enables the dashboard to group related papers cleanly and track macro-to-micro transitions smoothly.
- **Contributor Metadata:** Authors, institutional affiliations (categorized by Academia vs. Industry sectors), and countries to map institutional influence and concentration.

To augment frontier-AI context, optional supplementary data may be integrated from the Epoch AI Models Database and the Stanford AI Index, linking academic publication trends with real-world model scaling properties.

Overall, the dataset combines numerical, categorical, temporal, and spatial data, making it suitable for both advanced statistical analysis and machine-learning-based exploration (such as dimensionality reduction for topic clustering and time-series forecasting for research velocity).

4 Visualization Challenge

This dataset is non-trivial to visualize because it contains multiple interconnected dimensions, including publication growth over time, highly skewed citation distributions, overlapping research topics, institutional concentration, and shifting research dynamics across sectors.

The key design challenge is to avoid creating a static bibliometric dashboard. The dashboard must support a visual reasoning workflow: users first see the growth pattern, then test whether scientific impact keeps up, then inspect structural concentration across institutions and fields, and finally explore how different research pressure signals combine over time.

5 Dashboard Wireframe

**Note: The following mockups represent the initial wireframe design for the dashboard layout. These interface components, chart types, and structural placements are subject to modification and iterative refinement during the development process to optimize user experience and visual clarity.*

The proposed layout for the interactive Python Shiny dashboard is structured into four primary analytical tabs, presented sequentially from top to bottom to form a unified visual discovery workflow:

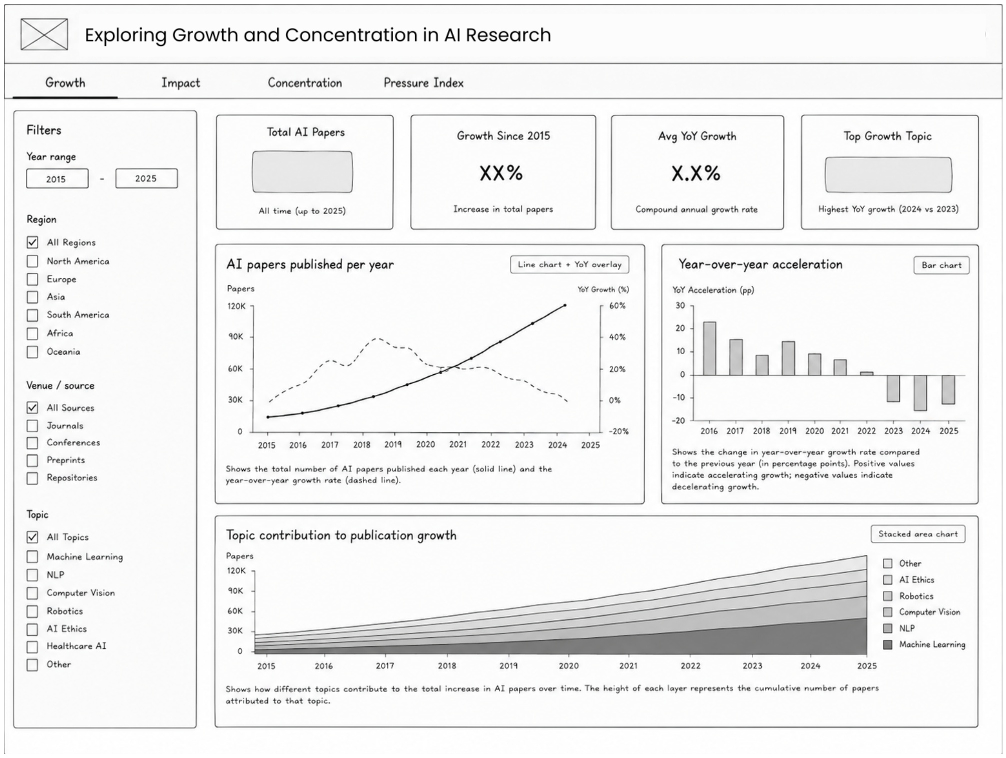

Figure 1: Publication Growth

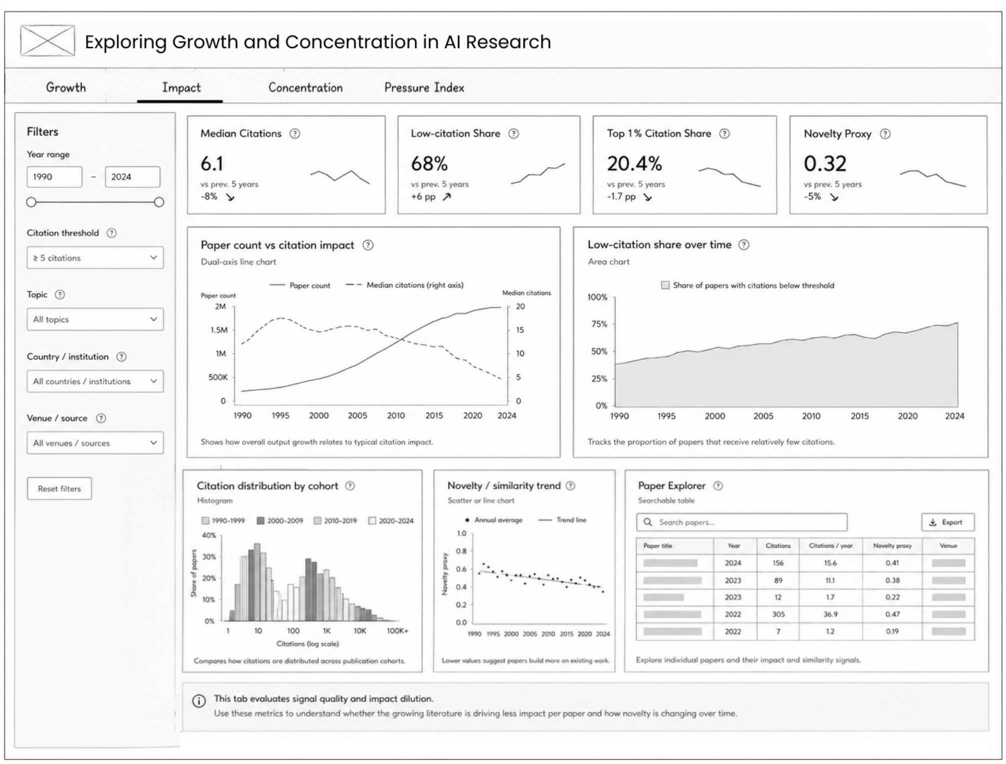

Figure 2: Research Impact

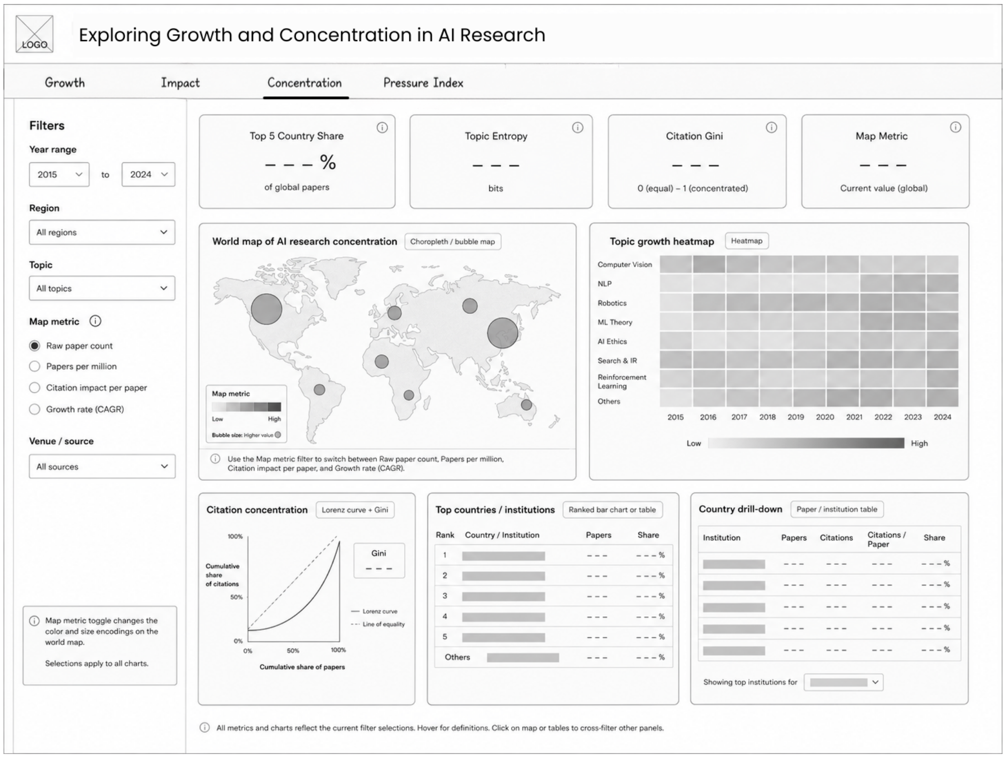

Figure 3: Concentration